

Ravi Shankar, PhD

Research Fellow | Clinical AI, Health Services Research, Health Informatics, Health Economics, Implementation Science, Qualitative Research, and Evidence Synthesis

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Professional Summary

Health services researcher working at the intersection of clinical artificial intelligence (AI), health economics and evidence synthesis. Currently Research Fellow at Tan Tock Seng Hospital, National Healthcare Group, and Visiting Research Clinician Scientist at the Rehabilitation Research Institute of Singapore, Nanyang Technological University. Research covers natural language processing (NLP) and large language models (LLMs) in healthcare, health technology assessment (HTA), economic evaluation, implementation science, systematic reviews and meta-analysis, digital health interventions and rehabilitation technology. PhD from the National University of Singapore; B.Tech and M.Tech from IIT Madras.

Key Achievements

- More than 100 publications: h-index 13, i10-index 18 (Google Scholar).
- First author in reputed journals, including npj Digital Medicine (IF 18.0), with original research published in Critical Care (IF 11.4), American Journal of Kidney Diseases (IF 9.4), Journal of Medical Internet Research (IF 8.2), and other leading journals.
- Principal Investigator on four competitive grants totalling SGD 452,563 (National Medical Research Council, NUHS, Synapse, National University of Singapore); SGD 672,663 in total research funding including one Co-Investigator grant.
- Research featured in The Straits Times (two articles) and by Synapse; NMRC Healthy Longevity Catalyst Award project listed among Healthy Longevity Challenge winners.

Skills and Areas of Expertise

- **Research areas:** Health services research; Health policy; Health technology assessment (HTA); Health economics; Implementation science; Digital health interventions
- **Clinical AI and data science:** Artificial intelligence (AI) in healthcare; Natural language processing (NLP); Large language models (LLMs); Clinical data extraction from health records; Sentiment analysis and topic modeling; Speech-based cognitive impairment detection; Clinical AI ethics and governance
- **Evidence synthesis:** Systematic reviews; Meta-analysis; Qualitative evidence synthesis; Systematic review protocols
- **Study design and evaluation:** Randomized controlled trial (RCT) design; Pilot and feasibility studies; Mixed-methods research; Qualitative research; Economic evaluation; Cost-effectiveness analysis
- **Digital and rehabilitation technology:** Rehabilitation robotics and exoskeletons; Telerehabilitation; Virtual reality (VR) interventions; Wearable sensors
- **Software:** Python; Stata; TreeAge Pro
- **Professional:** Grant writing; Project management

Professional Experience

Research Fellow

Tan Tock Seng Hospital, National Healthcare Group (NHG), Singapore
Clinical Research & Innovation Office (CRIO) | *October 2025 – Present*

Visiting Research Clinician Scientist

Nanyang Technological University (NTU), Singapore

Rehabilitation Research Institute of Singapore (RRIS) | *October 2025 – Present*

Research Fellow

Alexandra Hospital, National University Health System (NUHS), Singapore

Medical Affairs – Research Innovation & Enterprise | *May 2022 – September 2025*

Education

Doctor of Philosophy (PhD), Project Management

National University of Singapore (NUS), Singapore

August 2017 – December 2021

Bachelor of Technology (B.Tech) and Master of Technology (M.Tech)

Indian Institute of Technology (IIT) Madras, Chennai, India

August 2010 – May 2015

Research Funding

Total competitive funding: SGD 672,663 across five grants (Principal Investigator on four; Co-Investigator on one). Amounts are total project value, including direct costs, salary support and indirect research costs.

- **Principal Investigator:** Potential of Artificial Intelligence to Track and Predict Cognitive Impairment in Stroke Rehabilitation. National Medical Research Council (NMRC), Healthy Longevity Catalyst Awards (HLCA) 2024. **SGD 164,450.**
- **Principal Investigator:** Enhancing Psychosocial Well-Being of End-Stage Kidney Disease Caregivers Through Virtual Reality-Based Mindfulness Interventions. National University Health System (NUHS), NUHS Seed Fund Junior (March 2024 call). **SGD 140,820.**
- **Principal Investigator:** Development and Evaluation of an AI-Driven Nursing Shift Handover Assistant. Synapse, GenAlus Challenge. **SGD 98,093.**
- **Principal Investigator:** An AI-Driven Chatbot System (BOTANIC) for the Early Detection of Caregiver Burden in Caregivers of End-Stage Kidney Disease: A Feasibility Study. National University of Singapore (NUS), Academic Health Programme (AHP) Fund 2024. **SGD 49,200.**
- **Co-Investigator:** Comparing the level of agreement amongst a self-reported questionnaire, trained health coach evaluation, and sentiment analysis in classifying patients' readiness. National University Health System (NUHS), Health Services Research (HSR) Seed Grant (September 2023 call). **SGD 220,100.**

Publications

Publication metrics: More than 100 publications | average journal impact factor 4.6 | h-index 13 | i10-index 18 (Google Scholar).

Listed newest first; within each year by publication type, then by journal impact factor. Q = journal quartile; IF = journal impact factor.

1. **Shankar, R., Ong, R. C. L., Yee, E., Duraisamy, R., Ho, B. C. Y., Sim, R., Kuah, C. W. K., & Ong, P. L. (2026).** Feasibility, acceptability, and functional outcomes of a home-based exergaming telerehabilitation program for adults with functional disabilities: A multi-center single-arm pilot study. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-026-02055-x> (Q1, IF: 6.0)
2. **Shankar, R., Neoh, E. C., Chua, K. S. G., Tan, T. L., Tay, M. R. J., Kim, J. M., Aguirre-Ollinger, G., Kulkarni, S. R., & Ong, P. L. (2026).** Feasibility, safety, and acceptability of robotic-assisted rehabilitation for patients with knee osteoarthritis. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-73345-6> (Q1, IF: 4.9)

3. **Shankar, R., Lim, A., & Xu, Q. (2026).** Nurses' lived experiences of generative artificial intelligence-enabled shift handover innovation: A descriptive phenomenological study. *Journal of Advanced Nursing*. <https://doi.org/10.1111/jan.70629> (Q1, IF: 3.8)
4. Chua, K. S. G., **Shankar, R.**, Yam, M. G. J., Yee, E., Krishnan, R. R., Plunkett, T. K., Chong, W. B., Low, J. A. M., Ng, S. G., & Rao, J. (2026). Feasibility, safety and patient-reported outcomes of 3D-printed externally-applied cranial orthosis (PEACO) after decompressive craniectomy during inpatient rehabilitation: A retrospective cohort study. Manuscript submitted for publication, *Healthcare*. <https://doi.org/10.3390/healthcare4509210> (Q1, IF: 3.4)
5. Chandran, G., Kamsani, N. S., Tang, N., Lai, H. S., Ong, T., Cheng, S.-A., Lim, J. H., **Shankar, R.**, & Chew, E. (2026). Case report: Efficacy of transcutaneous spinal cord stimulation combined with overground robotic-assisted gait training in chronic incomplete spinal cord injury. *Frontiers in Human Neuroscience*. <https://doi.org/10.3389/fnhum.2026.1795068> (Q1, IF: 3.6)
6. **Shankar, R.**, Tan, C., Devi, F., Ang, E., & Er, J. (2026). Nurses' lived experiences managing peripherally inserted central catheters in low-volume settings: A phenomenological study of professional liminality. *The Journal of Nursing Research*. (Q1, IF: 2.7)
7. **Shankar, R.**, Tong, P., Yee, E., & Ong, P. L. (2026). Feasibility and usability of a wearable sensor-based telerehabilitation program for home-based exercise in patients with lower limb amputation: A pilot study. *Frontiers in Rehabilitation Sciences*. <https://doi.org/10.3389/fresc.2026.1828171> (Q1, IF: 2.5)
8. **Shankar, R.**, & Xu, Q. (2026). Mapping the discourse on neuropathic pain after spinal cord injury: A signal-generation natural language processing study of Reddit communities. *Spinal Cord*. <https://doi.org/10.1038/s41393-026-01286-y> (Q2, IF: 1.8)
9. **Shankar, R.**, Goh, A., & Xu, Q. (2026). Natural language processing-driven insights from social media: Topic modeling and sentiment analysis of healthcare sustainability discourse. *International Journal of Environmental Medicine*, 1(1), 4. <https://doi.org/10.3390/ijem1010004>
10. **Shankar, R.**, Lim, A., & Xu, Q. (2026). Understanding public discourse on Alzheimer's disease and dementia: A sentiment analysis and topic modeling study of social media data. *Journal of Dementia and Alzheimer's Disease*, 3(3), 34. <https://doi.org/10.3390/jdad3030034>
11. **Shankar, R.**, Sundar, V., Goh, Z., Sin, P. Y., & Tan, E. Y. (2026). Performance of large language models for extracting clinical data from breast cancer pathology reports: A systematic review. *npj Digital Medicine*. <https://doi.org/10.1038/s41746-026-02789-x> (Q1, IF: 18.0)
12. **Shankar, R.**, Bundeale, A., Yap, A., Goh, A., Rao, A., Low, J., & Yeo, S. C. (2026). Barriers and facilitators for the adoption of peritoneal dialysis: A systematic review of qualitative studies. *American Journal of Kidney Diseases*. <https://doi.org/10.1053/j.ajkd.2026.07.013> (Q1, IF: 9.4)
13. **Shankar, R.**, Devi, F., Kee, K., Low, E., & Ali, J. (2026). Experiences of stigma, bias, and communication challenges among pregnant healthcare workers: A systematic review of qualitative evidence. *International Journal of Nursing Studies*. <https://doi.org/10.1016/j.ijnurstu.2026.105629> (Q1, IF: 8.7)
14. **Shankar, R.**, Wang, L., Ho, S. H., Liew, M. F., Gollamudi, S. P. K., Wong, T. C., & Wong, S. (2026). Cost-effectiveness of virtual emergency care models: Systematic review. *JMIR mHealth and uHealth*, 14, e82143. <https://doi.org/10.2196/82143> (Q1, IF: 6.3)
15. **Shankar, R.**, Lim, A., & Xu, Q. (2026). Performance of large language models in data extraction for evidence synthesis: A systematic review. *Journal of Biomedical Informatics*. <https://doi.org/10.1016/j.jbi.2026.105086> (Q1, IF: 5.9)
16. **Shankar, R.**, Wang, L., Hoe, H. S., Lee, I. Y., Liew, M. F., Gollamudi, S. P. K., Wong, T. C., & Wong, S. (2026). The role of artificial intelligence in virtual emergency care: A systematic review. *International Journal of Medical Informatics*. <https://doi.org/10.1016/j.ijmedinf.2026.106411> (Q1, IF: 5.0)
17. **Shankar, R.**, Ng, Y. X., Teo, R. F., Yu, J., Devi, F., Foo, T. H., & Xu, Q. (2026). Determinants of family caregiver well-being in home-based end-of-life care: A qualitative synthesis. *Journal of Pain and Symptom Management*. <https://doi.org/10.1016/j.jpainsymman.2026.08.016> (Q1, IF: 4.0)

18. **Shankar, R., Goh, Z., & Xu, Q. (2026).** Techniques, performance, and feasibility of natural language processing for abstract screening in evidence synthesis: A systematic review. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261469813> (Q1, IF: 3.8)
19. **Shankar, R., Devi, F., Joshua, J. W., & Ng, T. M. (2026).** Barriers and enablers to pharmacist involvement in social prescribing: A systematic review of qualitative studies. *Research in Social and Administrative Pharmacy*. <https://doi.org/10.1016/j.sapharm.2026.01.005> (Q1, IF: 3.4)
20. **Shankar, R., Lim, A., & Xu, Q. (2026).** Barriers and facilitators to the use of large language model-based conversational agents in mental healthcare: A systematic review. *Healthcare*. <https://doi.org/10.3390/healthcare14101267> (Q1, IF: 3.4)
21. **Yap, E. Y. A., Chia, D. B., Lee, J. J.-M., Choo, M. H., Cheng, H. K., & Shankar, R. (2026).** The effectiveness of psychoeducational interventions for family caregivers of people with dementia in Asia: A systematic review. *Journal of Alzheimer's Disease*. <https://doi.org/10.1177/13872877261461174> (Q2, IF: 3.4)
22. **Shankar, R., Xu, Q., & Lim, A. (2026).** Leadership and psychological safety in healthcare: A qualitative evidence synthesis. *Leadership in Health Services*. <https://doi.org/10.1108/LHS-01-2026-0006> (Q2, IF: 2.3)
23. **Shankar, R., Lim, A., & Xu, Q. (2026).** Effectiveness of digital tools in supporting young caregivers: A systematic review. *Early Intervention in Psychiatry*. <https://doi.org/10.1111/eip.70215> (Q3, IF: 2.3)
24. **Shankar, R., Goh, Z., Xu, Q., Chew, E., & Chua, K. S. G. (2026).** Perceptions, attitudes, and lived experiences of therapists with lower limb robotic exoskeletons in stroke rehabilitation: A systematic review of qualitative studies. *Disability and Rehabilitation*. <https://doi.org/10.1080/09638288.2026.2637191> (Q2, IF: 2.2)
25. **Shankar, R., Tan, A., & Xu, Q. (2026).** Lived experiences of patients using left ventricular assist devices (LVADs): A systematic review of qualitative studies. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-01907-0> (Q4, IF: 1.3)
26. **Shankar, R., Xu, H., & Xu, Q. (2026).** Natural language processing and large language models for analyzing electronic health records and clinical notes in dementia research: A systematic review. *Journal of Dementia and Alzheimer's Disease*. <https://doi.org/10.3390/jdad3030034>
27. **Shankar, R., Kumar, F. D. S., Bundeale, A., & Mukhopadhyay, A. (2026).** Effectiveness and user experience of ambient artificial intelligence-driven scribe technology among clinicians: A protocol for a systematic review. *Systematic Reviews*. <https://doi.org/10.1186/s13643-026-03220-y> (Q1, IF: 5.7)
28. **Shankar, R., Bundeale, A., Low, J., & Hong, W. Z. (2026).** Barriers and facilitators to dietary adherence in adults with chronic kidney disease: A protocol for systematic review. *Systematic Reviews*. <https://doi.org/10.1186/s13643-026-03080-6> (Q1, IF: 5.7)
29. **Shankar, R., Chua, K., Krishnan, R. R., Tham, S.-L., Tan, L., Lai, A., Lau, M., Koh, Z. K., Thia, E., Shen, N., & Ong, P. L. (2026).** Digitization of Rehabilitation Evaluation and Activity Monitoring Using Thermography (DREAMT): Protocol for a mixed-methods randomized multiple baseline study in inpatient rehabilitation. *Frontiers in Digital Health*. <https://doi.org/10.3389/fdgth.2026.1884320> (Q1, IF: 5.2)
30. **Shankar, R., Shafawati, N., & Chandran, G. (2026).** Effectiveness of overground robotic exoskeletons in early inpatient rehabilitation for incomplete spinal cord injury: A systematic review protocol. *Frontiers in Neuroscience*. <https://doi.org/10.3389/fnins.2026.1781656> (Q2, IF: 4.0)
31. **Shankar, R., Tang, N., & Chandran, G. (2026).** Combined repetitive transcranial magnetic stimulation and transcutaneous spinal cord stimulation for upper limb recovery in chronic incomplete cervical spinal cord injury: Protocol for a pilot randomized controlled trial. *Frontiers in Neuroscience*. <https://doi.org/10.3389/fnins.2026.1833493> (Q2, IF: 4.0)
32. **Shankar, R., & Chandran, G. (2026).** Lower limb rehabilitation in chronic incomplete SCI: A randomized controlled trial protocol comparing combined TMS-tSCS neuromodulation versus tSCS alone. *Frontiers in Neuroscience*. <https://doi.org/10.3389/fnins.2026.1773917> (Q2, IF: 4.0)

33. **Shankar, R., Sundar, V., & Chua, K. S. G. (2026).** Barriers and facilitators to the adoption of speech-based cognitive screening tools: A protocol for systematic review of qualitative evidence. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261460855> (Q1, IF: 3.8)
34. **Shankar, R., Ang, L. L., & Xu, Q. (2026).** Effectiveness of virtual reality in supporting informal caregivers: A protocol for a systematic review and meta-analysis. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261486864> (Q1, IF: 3.8)
35. **Shankar, R., Devi, F., Xu, Q., & Lim, A. (2026).** Emerging roles and care impacts of geriatric pharmacists in outpatient multidisciplinary frailty and geriatric syndrome management: A systematic review protocol. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261482112> (Q1, IF: 3.8)
36. **Shankar, R., Sundar, V., & Tay, M. R. J. (2026).** Barriers and facilitators to uptake of non-surgical interventions for knee osteoarthritis: A protocol for a systematic review of qualitative studies. *Frontiers in Medicine*. <https://doi.org/10.3389/fmed.2026.1791068> (Q1, IF: 3.6)
37. **Shankar, R., Devi, F., & Xu, Q. P. (2026).** Effectiveness of digital tools in promoting advance care planning: A systematic review protocol. *Frontiers in Medicine*. <https://doi.org/10.3389/fmed.2025.1698865> (Q1, IF: 3.6)
38. **Shankar, R., Lo, Y. T., Fong, C. L., Keong, N., & Chua, K. S. G. (2026).** Motor imagery brain–computer interface (MI-BCI)-assisted upper limb neurorehabilitation for acute stroke during inpatient rehabilitation: A prospective feasibility study with economic evaluation protocol. *Journal of Clinical Medicine*. <https://doi.org/10.3390/jcm15145692> (Q1, IF: 3.3)
39. **Shankar, R., Choo, S. X., Chen, Z., Kuah, C. W. K., Plunkett, T. K., Ng, C. Y., Lin, S., Goh, K. H., Yee, E., Ge, X., Zhang, D., Chong, W. B., Low, J. A. M., Lau, M. S. E., Lim, X. Y., Naing, S. Y., Wong, L. T., Noronha, B., Aguirre-Ollinger, G., Hussain, A., Ong, P. L., & Chua, K. S. G. (2026).** Telerehabilitation Robotics for Upper Limb Rehabilitation after Stroke (TRUST): A multi-site pragmatic trial protocol. *Frontiers in Neurology*. <https://doi.org/10.3389/fneur.2026.1775829> (Q2, IF: 3.3)
40. **Shankar, R., Hue, Y. L., & Chua, K. S. G. (2026).** Neurosurgical intervention versus conservative management in older adults with traumatic brain injury: A protocol for a systematic review and meta-analysis. *Frontiers in Neurology*. <https://doi.org/10.3389/fneur.2026.1775791> (Q2, IF: 3.3)
41. **Shankar, R., Lim, A., & Xu, Q. (2026).** Emergency nurses' experiences with artificial intelligence–driven triage clinical decision support systems: A protocol for a systematic review of qualitative studies. *Journal of Emergency Nursing*. <https://doi.org/10.1016/j.jen.2026.05.010> (Q1, IF: 3.2)
42. **Shankar, R., Ong, P. L., & Chua, K. S. G. (2026).** Cost-effectiveness of home-based fall detection and monitoring technologies for community-dwelling older adults: A systematic review protocol. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2026-117122> (Q2, IF: 2.5)
43. **Shankar, R., Lim, A., & Xu, Q. (2026).** A qualitative systematic review protocol of patient experiences of using large language models for health information seeking and healthcare decision making. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-02067-x> (Q4, IF: 1.3)
44. **Shankar, R., Lim, A., & Xu, Q. (2026).** A systematic review protocol of qualitative studies on lived experiences of stigmatization in men with eating disorders. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-02230-4> (Q4, IF: 1.3)
45. **Shankar, R., Lim, A., & Xu, Q. (2026).** Cardiovascular and cerebrovascular safety of calcitonin gene-related peptide (CGRP)-targeted monoclonal antibodies and gepants for the preventive treatment of migraine in patients with pre-existing vascular risk factors: A systematic review and meta-analysis protocol. *SN Comprehensive Clinical Medicine*. <https://doi.org/10.1007/s42399-026-02446-0>
46. **Shankar, R. (2026).** The Jevons trap: When artificial intelligence in healthcare creates endless work. *The American Journal of Medicine*. <https://doi.org/10.1016/j.amjmed.2026.05.010> (Q1, IF: 6.0)
47. **Shankar, R. (2026).** Beyond the package insert: Operationalising lower clozapine doses for Asian patients in routine clinical practice. *Asian Journal of Psychiatry*. <https://doi.org/10.1016/j.ajp.2026.105036> (Q1, IF: 4.9)

48. **Shankar, R., & Xu, Q. (2026).** Advancing artificial intelligence in renal nutrition: The need for culturally adaptive, equity-focused frameworks beyond retrieval-augmented generation. *Journal of Renal Nutrition*. <https://doi.org/10.1053/j.jrn.2026.04.016> (Q1, IF: 4.0)
49. **Shankar, R. (2026).** Comment on 'Antenatal lifestyle interventions to reduce gestational weight gain: where should we start and how much will it cost?' *Public Health*. <https://doi.org/10.1016/j.puhe.2026.106422> (Q1, IF: 3.5)
50. **Shankar, R. (2026).** Large language models in post-acute and long-term care: Bridging promise and practice. *Journal of the American Medical Directors Association*. <https://doi.org/10.1016/j.jamda.2026.106379> (Q1, IF: 3.4)
51. **Shankar, R. (2026).** Navigating professional accountability in AI-assisted nursing practice: Ethical and legal imperatives for the digital age. *SAGE Open Nursing*. <https://doi.org/10.1177/23779608261457722> (Q1, IF: 2.9)
52. **Shankar, R. (2026).** Bridging implementation gaps: The critical role of institutional ecosystems in AI integration for nursing education. *SAGE Open Nursing*. <https://doi.org/10.1177/23779608261476555> (Q1, IF: 2.9)
53. **Shankar, R. (2026).** From cloud to pocket: Frontier edge foundation models and the future of clinical informatics. *Applied Clinical Informatics*. <https://doi.org/10.1055/a-2969-1166> (Q3, IF: 2.7)
54. **Shankar, R., & Xu, Q. (2026).** Beyond accuracy: Advancing LLM evaluation in clinical neurology toward real-world decision support. *Neurological Sciences*. <https://doi.org/10.1007/s10072-026-09125-x> (Q3, IF: 2.7)
55. **Shankar, R. (2026).** Beyond the standalone module: A case for retrofitting existing curricular infrastructure to embed planetary health in medical science education. *Medical Science Educator*. <https://doi.org/10.1007/s40670-026-02869-3> (Q2, IF: 1.8)
56. **Shankar, R. (2026).** Sovereign large language models for structured data extraction from pathology reports: A perspective for the clinical laboratory. *Laboratories*, 3(3), 9. <https://doi.org/10.3390/laboratories3030009>
57. **Shankar, R., Bundeale, A., & Mukhopadhyay, A. (2026).** Explainable artificial intelligence in speech-based cognitive decline detection: A systematic review. In J. M. Juarez et al. (Eds.), *Explainable Artificial Intelligence and Process Mining Applications for Healthcare (Communications in Computer and Information Science, Vol. 2749, pp. 40–55)*. Springer, Cham. https://doi.org/10.1007/978-3-032-21514-7_5
58. **Xu, P., Xu, J., Clark, J., & Shankar, R. (2026).** AI-enabled climate-health nexus framework for decision-support in sustainable urban green infrastructure. *Proceedings of the 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026)*, 896–900. <https://doi.org/10.22260/ISARC2026/0115>
59. **Shankar, R., & Mukhopadhyay, A. (2025).** Social media insights on sepsis management using advanced natural language processing techniques. *Critical Care*, 29(1), 115. <https://doi.org/10.1186/s13054-025-05344-4> (Q1, IF: 11.4)
60. **Shankar, R., Xu, Q., & Bundeale, A. (2025).** Patient voices in dialysis care: Sentiment analysis and topic modeling study of social media discourse. *Journal of Medical Internet Research*, 27, e70128. <https://doi.org/10.2196/70128> (Q1, IF: 8.2)
61. **Sumner, J., Shankar, R., Bundeale, A., Yap, A., Mohamed Ali, J., Teng, G. G., Phang, K. F., Yip, A. W., & Lim, Y. W. (2025).** Insights from high and low clinical users of telemedicine: A mixed-methods study of clinician workflows, sentiments, and user experiences. *International Journal of Medical Informatics*, 203, 106044. <https://doi.org/10.1016/j.ijmedinf.2025.106044> (Q1, IF: 5.0)
62. **Shankar, R., & Yip, A. (2025).** Sentiment analysis and topic modeling of social media data to explore public discourse on irritable bowel syndrome. *Scientific Reports*, 15(1), 10829. <https://doi.org/10.1038/s41598-025-08599-7> (Q1, IF: 4.9)

63. **Shankar, R., & Tan, W. (2025).** Strategic stakeholder management in public–private partnerships. *Engineering, Construction and Architectural Management*. <https://doi.org/10.1108/ECAM-12-2024-1681> (Q1, IF: 4.5)
64. **Shankar, R., Tang, N., Shafawati, N., Phan, P., Mukhopadhyay, A., & Chew, E. (2025).** Cost-effectiveness analysis of robotic exoskeleton versus conventional physiotherapy for stroke rehabilitation in Singapore from a health system perspective. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2024-095269> (Q2, IF: 2.5)
65. **Shankar, R., & Yip, A. (2025).** Transforming patient feedback into actionable insights through natural language processing: A knowledge discovery and action research study. *JMIR Formative Research*. <https://doi.org/10.2196/69699> (Q2, IF: 2.4)
66. **Shankar, R., Bundele, A., & Mukhopadhyay, A. (2025).** Natural language processing of electronic health records for early detection of cognitive decline: A systematic review. *npj Digital Medicine*, 8(1), 133. <https://doi.org/10.1038/s41746-025-01527-z> (Q1, IF: 18.0)
67. **Shankar, R., Kannan, J., Tan, Y. H., & Xu, Q. (2025).** Natural language processing techniques to detect delirium in hospitalized patients from clinical notes: A systematic review. *npj Digital Medicine*. <https://doi.org/10.1038/s41746-025-02051-w> (Q1, IF: 18.0)
68. **Shankar, R., Goh, Z., Devi, F., & Xu, Q. (2025).** A systematic review of explainable artificial intelligence methods for speech-based cognitive decline detection. *npj Digital Medicine*, 8, 724. <https://doi.org/10.1038/s41746-025-02105-z> (Q1, IF: 18.0)
69. **Shankar, R., Wen, K. S. W., Goh, Z., Tan, B., & Chandran, G. (2025).** Effectiveness of transcutaneous spinal cord stimulation for lower limb rehabilitation in spinal cord injury: A systematic review. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-025-01775-w> (Q1, IF: 6.0)
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